

DENSITY AND MORPHOLOGY OF DENDRITIC SPINES IN MOUSE NEOCORTEX

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Abstract—Dendritic spines of pyramidal cells are the main postsynaptic targets of cortical excitatory synapses and as such, they are fundamental both in neuronal plasticity and for the integration of excitatory inputs to pyramidal neurons. There is significant variation in the number and density of dendritic spines among pyramidal cells located in different cortical areas and species, especially in primates. This variation is believed to contribute to functional differences reported among cortical areas. In this study, we analyzed the density of dendritic spines in the motor, somatosensory and visuo-temporal regions of the mouse cerebral cortex. Over 17,000 individual spines on the basal dendrites of layer III pyramidal neurons were drawn and their morphologies compared among these cortical regions. In contrast to previous observations in primates, there was no significant difference in the density of spines along the dendrites of neurons in the mouse. However, systematic differences in spine dimensions (spine head size and spine neck length) were detected, whereby the largest spines were found in the motor region, followed by those in the somatosensory region and those in visuo-temporal region. © 2005 IBRO. Published by Elsevier Ltd. All rights reserved.

Key words: intracellular injection, Lucifer Yellow, dendritic spine, pyramidal neuron, spine density, dendrite.

Dendritic spines (herein referred to simply as spines) were first described by Santiago Ramón y Cajal who suggested that they were key elements in neuronal physiology that served to connect axons with dendrites (Cajal, 1888, 1890, 1911). This observation was confirmed several decades later by Gray (1959) who demonstrated by electron microscopy that spines established synapses, which led to a renewed interest in these structures. Spines are commonly characterized by the presence of a head and a neck, and although a variety of sizes and morphologies have been described, the most common subtypes are the pedunculated (thin and fungiform) and the stubby or sessile (Jones

and Powell, 1969; Peters and Kaiserman-Abramof, 1970; reviewed in Harris et al., 1992; Elston and DeFelipe, 2002). The size of the head and the length of the neck of spines appear to be important in determining their biophysical properties. For example, a direct relationship has been found between spine head size and the postsynaptic density, number of postsynaptic receptors, the number of pre-synaptic docked vesicles and the ready releasable pool of neurotransmitter. Similarly, the length of the neck has been related to calcium compartmentalization (Wilson et al., 1983; Harris and Stevens, 1989; Nusser et al., 1998; Schikorski and Stevens, 1999, 2001; Majewska et al., 2000a,b; Sabatini et al., 2001; Yuste and Majewska, 2001; Segal, 2002; Holthoff et al., 2002; Yuste and Bonhoeffer, 2001, 2004; Noguchi et al., 2005).

Differences between cortical areas and species have been identified in the complexity of dendritic trees and in the density, number and distribution of spines along the dendrites of pyramidal neurons, particularly in the neocortex of primates (e.g. Elston et al., 2001, 2005; Jacobs et al., 2001; Elston and Rockland, 2002; Elston, 2003), which are likely to influence synaptic and neuronal function (reviewed in Elston, 2002, 2003; Jacobs and Scheibel, 2002). Furthermore, differences in the complexity of the basal dendritic trees and in spine morphology of pyramidal neurons have recently been demonstrated between different cortical layers and regions in the mouse (Benavides-Piccione et al., 2002, 2005; Konur et al., 2003). However, there is little information available regarding the morphology, density and distribution of spines in different cortical areas of the mouse neocortex. As rodents are the most commonly used model in physiological experiments of cortical function and mental illness, and the results of these experiments are transferred to human function and illness, it is important to determine whether or not pyramidal cell structure varies among cortical areas in the mouse neocortex and, if so, the extent to which phenotypic variation in neuron structure subserves cortical function.

In the present study, we used intracellular injection of Lucifer Yellow in fixed tissue and light-microscopic reconstruction to analyze the density and distribution of spines in the motor (M2), somatosensory (S2) and visuo-temporal (V2L/TeA) cortex of the mouse. In addition, we compared the morphology of spines between M2, S2, and V2L/TeA. In contrast to primates, there were no significant differences in spine density in the basal dendrites of pyramidal neurons among cortical areas. However, we found significant differences in spine head sizes and neck length between these regions.

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Abbreviations: DAB, 3,3'-diaminobenzidine; M2, motor cortex; S2, somatosensory cortex; V2L/TeA, visuo-temporal cortex.

EXPERIMENTAL PROCEDURES

Preparation of material and cell injections

C57BL/6 mice ($n=2$ males, 2 months old) were anesthetized by administering a lethal i.p. injection of sodium pentobarbitone and they were then perfused intracardially with 4% paraformaldehyde in 0.1 M phosphate buffer. All experiments were performed in accordance with the guidelines established by the European Union regarding the use and care of laboratory animals. The brains were removed from each animal and the cortex of the right hemisphere was flattened between two glass slides before immersing it in 4% paraformaldehyde for a further 24 h (Welker and

Woolsey, 1974). The cerebral cortex was cut with the aid of a Vibratome (Vibratome, St. Louis, MO, USA) and sections were obtained tangential to the cortical surface. Intracellular injections were performed according to the methods described in earlier studies (Elston and Rosa, 1997; Elston, 2001). Briefly, cells in the M2, S2 and V2L/TeA cortex of the mice (areas M2, S2 and V2L/TeA of Franklin and Paxinos, 1997) were injected individually with Lucifer Yellow by applying continuous current until the distal tips of each dendrite fluoresced brightly. After performing the injections, the sections were incubated with an antibody to Lucifer Yellow diluted 1:100,000 in stock solution (2% bovine serum albumin [Sigma A3425], 1% Triton X-100 [BDH 30632], 5% su-

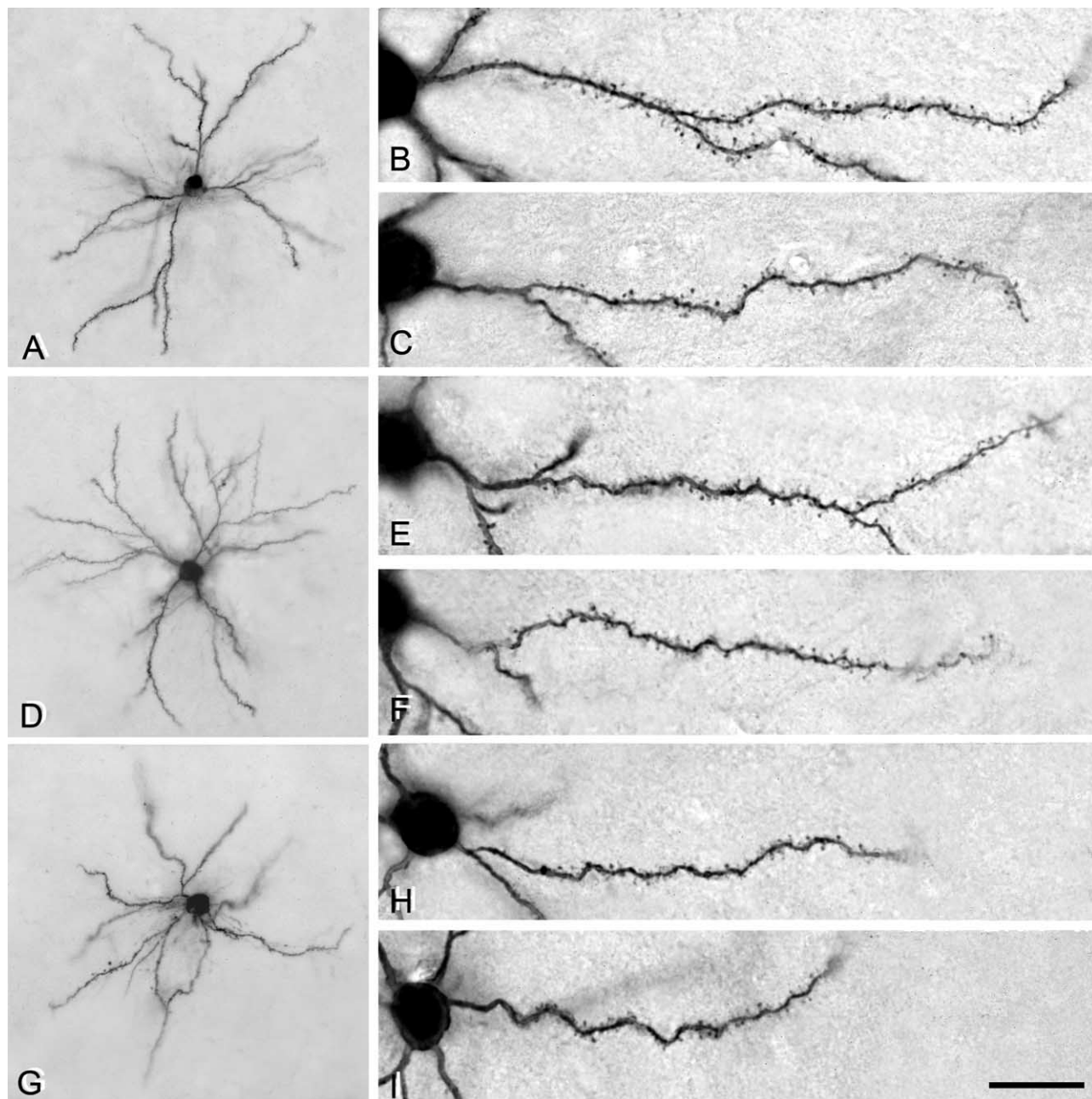


Fig. 1. Photomicrographs of pyramidal cells of layer III injected with Lucifer Yellow and processed with DAB. The cells illustrated were sampled from the mouse neocortical regions M2 (A–C), S2 (D–F) and V2L/TeA (G–I). (B, C, E, F, H, I) Examples of horizontally projecting dendrites. Scale bar=60 μ m in A, D, G and 20 μ m in B, C, E, F, H, I.

crose in 0.1 M phosphate buffer). Subsequently, the antibody was detected with a specific biotinylated secondary antibody (Amersham RPN 1004; 1:200 in stock solution), which was visualized with a biotin–horseradish peroxidase complex (Amersham RPN1051; 1:200 in 0.1 M phosphate buffer) and the DAB chromogen (3,3'-diaminobenzidine; Sigma D 8001) (Fig. 1).

Reconstruction and analysis

Sholl analysis. Ninety pyramidal cells (15 cells per area and per mouse) were reconstructed to perform the Sholl analysis, and to quantify the branching pattern of neurons in the different cortical areas (Sholl, 1953) in both two- and three-dimensions (2-D and 3-D, respectively). 2-D analysis was performed by counting the number of dendritic branches that intersected successive concentric circles (25 μm increments in radii) for each cell. 3-D analysis was carried out using Neurolucida (MicroBrightfield, Inc., Williston, VT, USA), by counting the number of dendritic branches that intersected successive concentric spheres (25 μm increments in radii) for each cell.

Density and total number of spines. To calculate the density of spines, 90 horizontally projecting basal dendrites (15 per cortical area and per mouse) from different cells were drawn at high power (100 $\times$ oil immersion objective) at a final magnification of 1600 $\times$. All the spines were drawn, including the sessile and pedunculate types, along the entire length of the dendrite, from the cell body to the distal tip of dendrites (Jones and Powell, 1969). No correction factors were applied to the spine counts (e.g. see Feldman and Peters, 1979; Larkman, 1991) as the high power reconstruction of cells permits all dendritic spines to be visualized (i.e., the DAB reaction product is more transparent than the Golgi reaction product).

The estimation of the total number of spines in the pyramidal cell basal dendritic tree was calculated by multiplying the average number of spines of a given portion of the dendrite by the average number of branches for the corresponding region over the entire dendritic tree (Elston, 2001), in both 2-D and 3-D.

Spine morphology. The methodology used for spine reconstruction and analysis has been described in detail previously (Benavides-Piccione et al., 2002). Briefly, we selected basal dendritic segments from 40 horizontally projecting basal dendrites of different cells in each area. The lateral spines on these dendrites were reconstructed to perform a morphometric analysis. The segments analyzed (30 μm long) corresponded to the regions of highest spine density in mice, beginning at 45 μm from the soma (Benavides-Piccione et al., 2002). Images were captured at different focal planes in the region of interest using a BX51 Olympus microscope (100 $\times$ objective) attached to Olympus DP50 camera (final magnification of 2800 $\times$). Thereafter, images were used to make composite projection drawings of the dendritic spines (Fig. 2D). Spine morphology, including the area of the head, the major and minor axis of the head, and the length of the neck, were quantified with both a digitizing tablet (SummaSketch III) and NIH image software (NIH Research Services, Bethesda, MD, USA). Stubby spines included also those spines whose head was not distinguishable from the neck.

RESULTS

Density and total number of spines on the basal dendrites

We first compared the structure of dendritic trees between different cortical areas. Plotting the results of the Sholl 2-D analysis and the 3-D analysis revealed that on average, pyramidal cell dendrites in M2 had more branches than those in S2. Furthermore, on average those in S2 had more branches than those in V2L/TeA (Fig. 3A). Similarly, the peak branching complexity in the basal dendritic trees of pyramidal

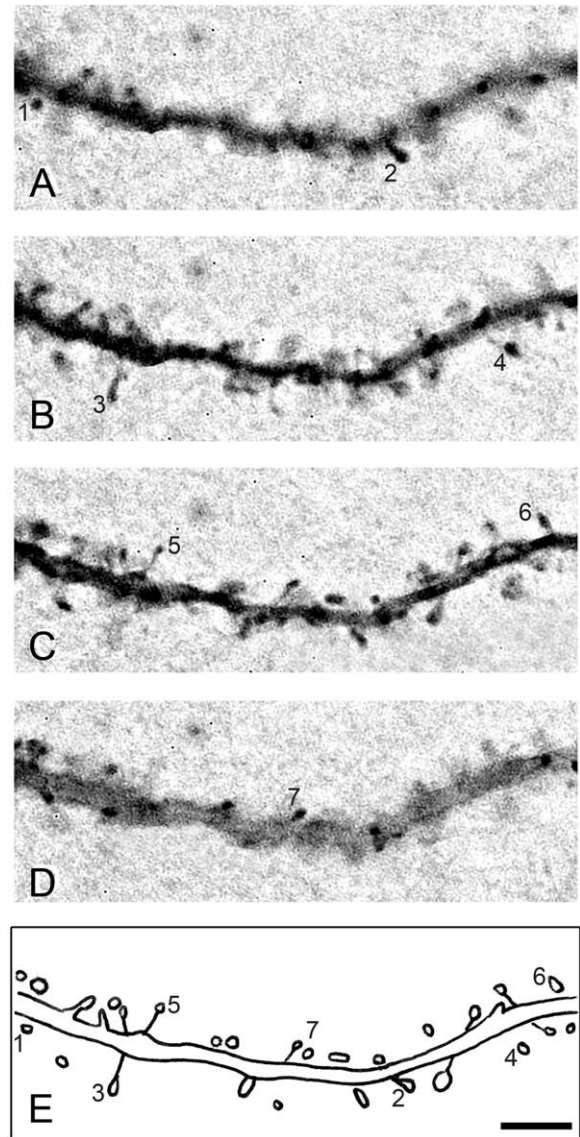


Fig. 2. Methods. (A–D) Photomicrographs of different focal planes of the same basal dendritic segment of a pyramidal neuron in M2. Note the appearance of spines in different focal planes. (E) Drawing showing the reconstruction of images illustrated in A–D, indicating some spines (1–7). Scale bar=3.75 μm .

cells in M2 was higher than that in S2 and V2L/TeA. An analysis of variance of the whole dendritic arbor as a function of the distance from the cell body to the distal tips revealed significant differences among the three cortical regions (2-D: $F_{2, 87}=99.25$; $P<0.001$; 3-D: $F_{2, 87}=94.71$; $P<0.001$). Moreover, there were no significant differences in the number of branches when compared between 2-D and 3-D Sholl analysis (M2: $F_{1, 58}=2.17$, $P>0.1$; S2: $F_{1, 58}=0.31$, $P>0.5$; V2L/TeA: $F_{1, 58}=3.8$, $P>0.05$).

Over 17,000 individual spines were drawn along 90 horizontally projecting basal dendrites of neurons in M2, S2 and V2L/TeA (6536, 5561 and 4857, respectively). Plots of the spine density as a function of the distance from

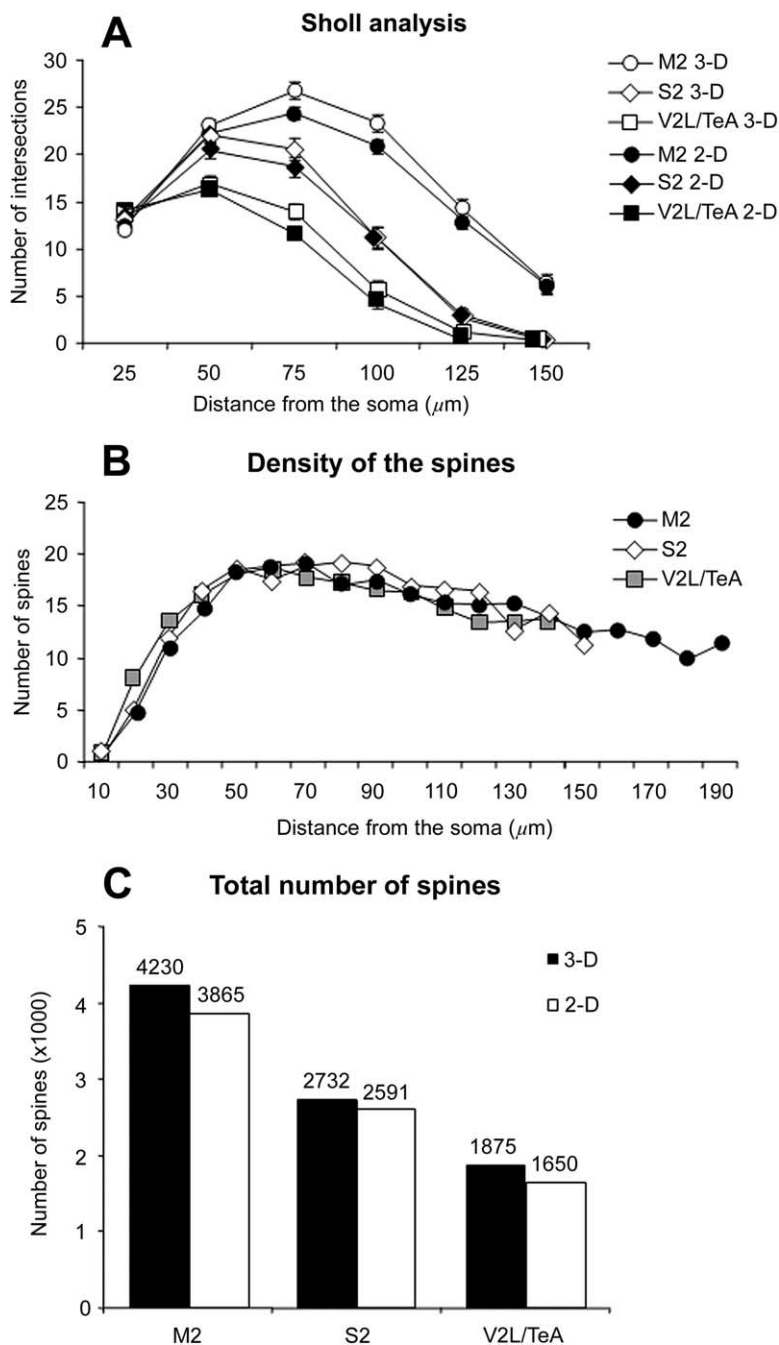


Fig. 3. (A) Plot of the number of intersections of the basal dendritic trees calculated in 3-D with Neurolucida and in 2-D by the classical Sholl analysis. (B) Plot of spine density calculated as the number of spines per $10 \mu\text{m}$ segment of the dendrite from the soma to the tip of the dendrite in M2, S2 and V2L/TeA. (C) Plot of the total number of spines estimated in 3-D and 2-D.

the cell body to the distal tips of the dendrites (per $10 \mu\text{m}$) are illustrated in Fig. 3B. Repeated ANOVA tests revealed no significant differences in the distribution of spines ($F_{2, 87}=0.603$, $P>0.5$) among cortical areas.

Combining the data from the 2-D Sholl analysis with that of the spine densities revealed that, on average, cells in the M2 had considerably more spines in their basal dendritic tree (3865) than those in the S2 (2591) and V2L/TeA (1650) (Fig. 3C). When these values were com-

pared with those obtained with the 3-D Sholl analyses, a similar trend with slightly higher values was revealed in each cortical area (M2, 4230; S2, 2732; V2L/TeA, 1875; Fig. 3C).

Spine morphology

In order to study the possible differences in spine morphologies in the mouse neocortex, over 956 individual spines

Table 1. Morphometric values (means $\pm$ S.E.M.) of dendritic spines of layer III pyramidal cells sampled in mouse M2, S2 and V2L/TeA

Parameter	V2L/TeA	S2	M2
Area of the head (μm^2)	0.31 $\pm$ 0.01 (<i>n</i> =1226)	0.37 $\pm$ 0.01 (<i>n</i> =1306)	0.58 $\pm$ 0.01 (<i>n</i> =956)
Major axis (μm)	0.77 $\pm$ 0.01 (<i>n</i> =1226)	0.85 $\pm$ 0.01 (<i>n</i> =1306)	1.09 $\pm$ 0.02 (<i>n</i> =956)
Minor axis (μm)	0.48 $\pm$ 0.01 (<i>n</i> =1226)	0.53 $\pm$ 0.01 (<i>n</i> =1306)	0.65 $\pm$ 0.02 (<i>n</i> =956)
Length of the neck (μm)	0.67 $\pm$ 0.01 (<i>n</i> =1226)	0.73 $\pm$ 0.01 (<i>n</i> =1306)	0.77 $\pm$ 0.02 (<i>n</i> =956)

Data from S2 and V2L/TeA cortex were taken from Benavides-Piccone, et al. (2002).

were drawn along 40 horizontally projecting basal dendrites of neurons in M2. The information extracted from these drawings was compared with previous results reported in S2 (*n*=1306) and V2L/TeA (*n*=1226) of the same animals (see Benavides-Piccone et al., 2002). The mean area of the spine heads in M2 was $0.58\pm0.01 \mu\text{m}^2$ (mean $\pm$ S.E.M.), higher than that observed in S2 and V2L/TeA (Table 1). Statistical analysis showed these differences to be significant (one-way ANOVA, $F_{2, 3485}=395.1$, $P<0.001$). The peak of the histogram showing the relative frequency of spine heads in M2 was skewed to the right (around $0.6 \mu\text{m}^2$) when compared with S2 and V2L/TeA (Fig. 4A). In addition, the necks of spines in M2 ($0.77\pm0.02 \mu\text{m}$) were significantly longer than those in

V2L/TeA (one-way ANOVA, $F_{2, 2151}=7.5$, $P<0.001$; post hoc Bonferroni analysis, $P<0.001$). However, there was no significant difference in the length of the spine necks between M2 and S2 (post hoc Bonferroni analysis, $P=0.35$; Table 1). Interestingly, the distribution of neck length in M2 was clearly bimodal, as in the S2 and V2L/TeA regions, with a peak at zero and a second peak at around $0.5 \mu\text{m}$. However, the percentage of spines considered to be without a neck was lower in M2 than in S2 or V2L/TeA (Fig. 4B).

Finally, we quantified the length of the major and minor axis of the spine heads to determine whether these parameters might differ between cortical areas. The spine heads in M2 had a mean major axis significantly greater than those observed in S2 and V2L/TeA (Table 1), major axis: one-way ANOVA, $F_{2, 3485}=239.32$, $P<0.001$; minor axis: one-way ANOVA, $F_{2, 3485}=321.31$, $P<0.001$). Post hoc comparisons revealed significant differences among all three cortical areas ($P<0.001$).

DISCUSSION

This study has generated two major findings. Firstly, we found no significant differences in spine density at the corresponding distances along the basal dendrites of layer III pyramidal cells between the M2, S2 and V2L/TeA regions of mice. In contrast, we did observe appreciable differences in our estimates of the total number of spines in the basal dendritic trees of pyramidal neurons between these three regions. Secondly, we found significant differences in the head size and neck length of spines in M2 when compared with other cortical regions.

It has been hypothesized that a link may exist between the number of spines in the dendritic trees of neocortical pyramidal neurons and the complexity of cortical processing. Indeed, it is believed that convergence subserves “higher” processing, which can be achieved by the integration of more inputs per neuron (Elston, 2000; Jacobs et al., 2001; Duan et al., 2002; see also Elston, 2002; Sala, 2002; Jacobs and Scheibel, 2002; Treves, 2005 for reviews). An increase in the number of inputs received by individual neurons could be achieved either by increasing the density of the spines along their dendrites and/or by increasing the length/number of dendrites. Our results demonstrate that in the mouse cortex, the spine density along the basal dendrites of pyramidal cells is similar in M2, S2 and V2L/TeA. This was an unexpected finding given the enormous variation in the density of spines seen between cortical areas in primates: up to a 30-fold difference has been reported in the number of spines in the dendritic trees of pyramidal cells in primates (reviewed in Elston and DeFelipe 2002). Furthermore, the peak spine density reported along the basal dendrites of layer III pyramidal cells in the mouse cortex (approximately 20 spines/10 μm) was higher than that of some occipital and temporal cortical areas in primates, including humans (e.g. Elston et al., 2001). For example, in macaque area 17 and human area 18 the peak spine density is approximately seven and 12 spines/10 μm , respectively. In contrast, the peak spine density in

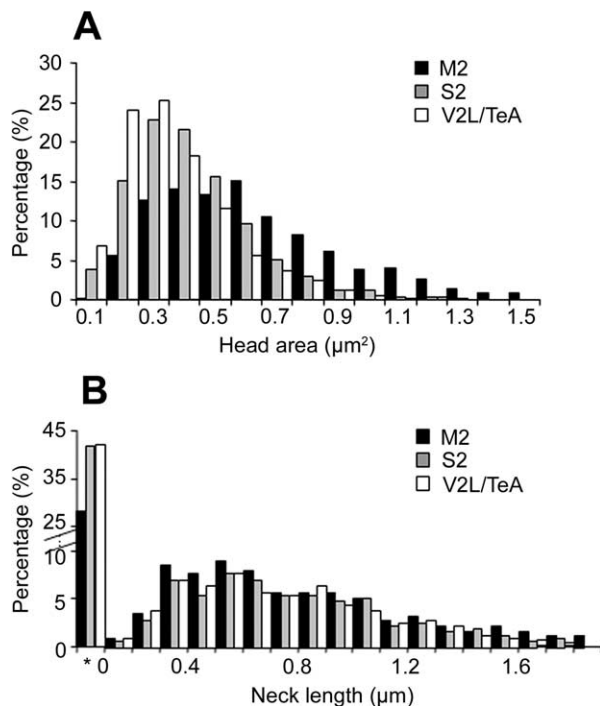


Fig. 4. Histograms of the relative frequency of spines. (A) Distribution of the spines as a function of the head area. (B) Distribution of the spines as a function of the neck length. M2 in black, S2 in dark gray and V2L/TeA in white. Data from S2 and V2L/TeA cortex were taken from Benavides-Piccone et al. (2002).

the mouse was considerably lower than that reported in the granular prefrontal cortex of the macaque and human (24 and 32 spines/10 μm , respectively; Elston et al., 2001). As in primates, the branching complexity and length of the basal dendrites were different in each cortical region of the mouse. Indeed, the branching complexity was greater in the M2 than in the S2, followed by the V2L/TeA (see also Benavides-Piccione et al., 2005). These structural variations in the dendritic trees resulted in the number of spines in M2 being 55% higher than in S2, and 125% higher than in V2L/TeA. Likewise, neurons in S2 had 46% more spines than those in V2L/TeA. Nevertheless, the estimated differences of the total number of spines in the basal dendritic tree of the “average” neuron between cortical regions were greater in primates than in the mouse. For example, in the granular prefrontal cortex of the human and macaque, the number of spines in the basal dendritic tree of layer III pyramidal cells is up to 16 and 23 times greater than in the primary visual cortex (see Elston, 2003 for a review). It remains to be determined whether the relative differences in the structure of pyramidal cells in the murine and primate cortex reflect the cortical areas included for study, the size of their brains or their phylogeny. The functional implications of area and species differences in dendritic morphology have been reviewed in detail elsewhere (Elston, 2002, 2006; Jacobs and Scheibel, 2002).

The analysis of the morphology of the dendritic spines in area M2 revealed that they have larger heads and longer necks than those in S2 and V2L/TeA. These morphological differences may influence various functions. For example, larger spine heads have been correlated with more postsynaptic receptors (Nusser et al., 1998), as well as more docked vesicles and a bigger readily releasable pool in presynaptic terminals (Schikorski and Stevens, 1999, 2001). Furthermore, neck length is one of the factors that control the rate of calcium decay as it diffuses from the synapse to the dendrite. Differences in the length of the spine necks therefore alter the calcium dynamics in spines, which in turn has been related to learning (Yuste and Bonhoeffer, 2001, 2004; Segal, 2002; Yuste and Majewska, 2001). Thus, the different morphology of spines in M2 might produce specific modulations of the synaptic current—different from S2 and V2L/TeA. Finally, it has recently been reported that small spines are preferential sites for long-term potentiation (learning spines), whereas large spines might represent physical traces of long-term memory (memory spines) (Matsuzaki et al., 2001, 2004; Kasai et al., 2003). This might indicate that the motor region in the mouse stores information that is more heavily protected from LTP and more resistant to plasticity than in other cortical areas. This may be a feature that is important for the memorization of motor tasks.

Caveats

There are two limitations in the present study that should be highlighted. Firstly, we only studied the basal dendritic tree of layer II/III pyramidal cells, for technical and comparative reasons. The basal dendritic tree was chosen due to methodological limitations which constrained us to

choose between either basal or apical dendrites (otherwise the dendrites are truncated). Layer II/III pyramidal cells were selected because the vast majority of quantitative studies of pyramidal cell structure in the cerebral cortex have been performed on the basal dendritic trees of layer III neurons in different primate species and cortical areas. Thus, we used the same methodology to facilitate the comparison of the information obtained with that established in other species, and to determine how the phenotypes of pyramidal neurons vary in different cortical areas and species. Therefore, while the strength of our approach is that it is highly focused, this restricts the general application of the observations and we cannot rule out that other samples may have led us to reach different conclusions.

Secondly, regarding the morphology of spines, we only reconstructed spines that protruded laterally from the dendritic tree, by neglecting the spines that were located directly above or below the dendrite. However, the spine head area and neck length should not be affected by our decision to reconstruct only lateral spines. In conclusion, the present results clearly show different attributes of spine morphology and in the number of spines on pyramidal neurons in the three mouse cortical regions examined. Nevertheless, to obtain a more global picture of the differences in pyramidal cell structure throughout the various cortical areas, further studies along these lines should include more comprehensive quantitative analysis of pyramidal cells. These studies must not only take into account their basal dendritic trees in the rest of cortical areas, but also throughout the cortical layers within each area. Their apical dendritic trees should also be considered.

Acknowledgments—We thank the members of our laboratories for their assistance, especially to J. Arellano for the discussions regarding statistics. This work was supported by the Spanish Ministry of Education and Science (BFI2003-02745), the Comunidad de Madrid (grant 08.5/0027/2001), the National Health and Medical Research Council of Australia (G.N.E.) and the National Eye Institute and the Human Frontiers Science Project (R.Y.). I.B.-Y. is research fellowship of the Spanish Ministry of Education and Science (AP 2001-0671).

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